Derivatives and Taylor Series

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Derivatives

Limits and derivatives

Definition 0.1 (Limit of a function at a point)**.** Let f be a function defined at every point of an open interval around c, except possibly at c itself, and let L be a real number. The **limit** of $f(x)$ as x approaches c is L, written $\lim_{x \to c} f(x) = L$, if for every $\varepsilon > 0$ there is a $\delta > 0$ such that

$$|f(x) - L| < \varepsilon \quad \text{for every } x \text{ with } 0 < |x - c| < \delta.$$

Here $|\cdot|$ is the absolute value. When such a real number L exists, the limit **exists**; otherwise, the limit does not exist. The value $f(c)$, if it is defined, plays no role.

Example 0.1 (The limit of $3x + 1$ at 2)**.** $\lim_{x \to 2}(3x + 1) = 7$. Given $\varepsilon > 0$, take $\delta = \varepsilon/3$. For every x with $0 < |x - 2| < \delta$, using the distributive law in the second line,

$$\begin{aligned} |(3x + 1) - 7| &= |3x - 6| && \text{(subtract)} \\ &= |3(x - 2)| && \text{(distributive law)} \\ &= 3|x - 2| && (|3y| = 3|y|, \text{ since } 3 > 0) \\ &< 3 \cdot \frac{\varepsilon}{3} && (|x - 2| < \delta = \varepsilon/3) \\ &= \varepsilon && \text{(multiply)} \end{aligned}$$

For example, with $\varepsilon = 0.3$ and $\delta = 0.1$, the point $x = 2.05$ satisfies $0 < |2.05 - 2| < 0.1$, and

$$\begin{aligned} |(3 \cdot 2.05 + 1) - 7| &= |7.15 - 7| \\ &= 0.15 \\ &< 0.3. \end{aligned}$$

Definition 0.2 (Right-hand limit)**.** Let f be a function and L a real number. The **right-hand limit** of f at c is L, written $\lim_{x \to c^+} f(x) = L$, if for every $\varepsilon > 0$ there is a $\delta > 0$ such that $|f(x) - L| < \varepsilon$ for every x with $c < x < c + \delta$.

Definition 0.3 (Left-hand limit)**.** Let f be a function and L a real number. The **left-hand limit** of f at c is L, written $\lim_{x \to c^-} f(x) = L$, if for every $\varepsilon > 0$ there is a $\delta > 0$ such that $|f(x) - L| < \varepsilon$ for every x with $c - \delta < x < c$.

Definition 0.4 (One-sided limits)**.** Let f be a function. The right-hand limit (Definition 0.2) and the left-hand limit (Definition 0.3) of f at c are the **one-sided limits** of f at c.

Example 0.2 (One-sided limits of a step function)**.** Let $H(x) = 1$ for $x \geq 0$ and $H(x) = 0$ for $x < 0$.

- $\lim_{x \to 0^+} H(x) = 1$, because $H(x) = 1$ for every $x > 0$.
- $\lim_{x \to 0^-} H(x) = 0$, because $H(x) = 0$ for every $x < 0$.

The one-sided limits differ, since $1 \neq 0$, so $\lim_{x \to 0} H(x)$ does not exist.

Remark (A limit exists exactly when both one-sided limits exist and are equal)

Remark 0.1 (A limit exists exactly when both one-sided limits exist and are equal)*.* The limit $\lim_{x \to c} f(x)$ (Definition 0.1) exists exactly when both one-sided limits (Definition 0.4) exist and are equal, and then all three are equal.

Videos

Jon Krohn's "Calculus for Machine Learning" YouTube playlist introduces limits and derivatives:

- Calculating Limits
- Exercises on Limits
- Intro to Differential Calculus
- How Derivatives Arise from Limits
- Derivative Notation

Definition 0.5 (Difference quotient)**.** Let f be a real-valued function of one variable, defined at c and at $c + h$, with $h \neq 0$. The **difference quotient** of f at c with increment

h is

$$\frac{f(c+h)-f(c)}{h},$$

the slope of the line through the points $(c, f(c))$ and $(c+h, f(c+h))$.

Example 0.3 (The difference quotient of x^2 at 1)**.** For $f(x) = x^2$, $c = 1$, and $h \neq 0$,

$$\begin{aligned}
\frac{f(1+h)-f(1)}{h} &= \frac{(1+h)^2-1}{h} && \text{(substitute into } f\text{)} \\
&= \frac{1+2h+h^2-1}{h} && \text{(expand } (1+h)^2\text{)} \\
&= \frac{2h+h^2}{h} && (1-1=0) \\
&= 2+h, && \text{(divide by } h \neq 0\text{)}
\end{aligned}$$

which tends to 2 as $h \to 0$ (Definition 0.1). With $h = 0.1$ the difference quotient is $2 + 0.1 = 2.1$: the line through $(1, 1)$ and $(1.1, 1.21)$ has slope

$$\begin{aligned}
\tfrac{1.21-1}{0.1} &= \tfrac{0.21}{0.1} \\
&= 2.1.
\end{aligned}$$

Definition 0.6 (Differentiable function)**.** A function f is **differentiable at** $x = c$ if the limit (Definition 0.1) of the difference quotient (Definition 0.5) as $h \to 0$,

$$f'(c) \stackrel{\text{def}}{=} \lim_{h \to 0} \frac{f(c+h)-f(c)}{h},$$

exists and is finite. The number $f'(c)$ is the derivative of f at c (Definition 0.7). (Larson and Edwards 2018, sec. 2.1, p. 100)

Example 0.4 (A differentiable function, and a limit that is not finite)**.**

- $f(x) = x^2$ is differentiable at $c = 3$:

$$\begin{aligned}
\frac{f(3+h)-f(3)}{h} &= \frac{(3+h)^2-3^2}{h} && \text{(substitute } f(x)=x^2\text{)}\\
&= \frac{9+6h+h^2-9}{h} && \text{(expand } (3+h)^2\text{)}\\
&= \frac{6h+h^2}{h} && \text{(cancel } 9-9\text{)}\\
&= 6+h && \text{(divide by } h\neq 0\text{)}
\end{aligned}$$

which tends to the finite limit $f'(3) = 6$ as $h \to 0$.

- $g(x) = \sqrt[3]{x}$ is not differentiable at $c = 0$: the difference quotient

$$\begin{aligned}
\frac{g(h)-g(0)}{h} &= \frac{\sqrt[3]{h}}{h}\\
&= \frac{1}{(\sqrt[3]{h})^2}
\end{aligned}$$

grows without bound as $h \to 0$, so the limit is not finite.

Definition 0.7 (Derivative)**.** If f is differentiable at c (Definition 0.6), the number $f'(c)$ is the **derivative** of f at c. The **derivative** of f is the function f' whose value at each x where f is differentiable is $f'(x)$.

Definition 0.8 (Lagrange's notation)**.** Writing the derivative (Definition 0.7) with a prime, as f' or $f'(x)$, is **Lagrange's notation**.

Definition 0.9 (Leibniz notation)**.** The derivative of f (Definition 0.7) is also written $\frac{df}{dx}$ or $\frac{d}{dx}f(x)$, which is **Leibniz notation**.

Example 0.5 (The derivative of x^2)**.** For $f(x) = x^2$, the computation in Example 0.4, with 3 replaced by any real number c, gives $\frac{f(c+h)-f(c)}{h} = 2c + h$, which tends to $2c$ as $h \to 0$. So the derivative of f is $f'(x) = 2x$. For example, $f'(3) = 6$ and $f'(-1) = -2$.

Definition 0.10 (Right-hand derivative)**.** Let f be a function defined at c. The **right-hand derivative** of f at c is the right-hand limit (Definition 0.2) $\lim_{h\to 0^+} \frac{f(c+h)-f(c)}{h}$ of the difference quotient (Definition 0.5), when that limit exists and is finite.

Definition 0.11 (Left-hand derivative)**.** Let f be a function defined at c. The **left-hand derivative** of f at c is the left-hand limit (Definition 0.3) $\lim_{h\to 0^-} \frac{f(c+h)-f(c)}{h}$, when that

limit exists and is finite.

Definition 0.12 (One-sided derivatives)**.** Let f be a function defined at c. The right-hand derivative (Definition 0.10) and the left-hand derivative (Definition 0.11) of f at c are the **one-sided derivatives** of f at c.

Example 0.6 (One-sided derivatives of $|x|$ at 0)**.** Let $f(x) = |x|$, the absolute value, and $c = 0$.

- For $h > 0$, $|h| = h$, so

$$\begin{aligned}\frac{f(0+h) - f(0)}{h} &= \frac{h - 0}{h} \\ &= 1.\end{aligned}$$

  The right-hand derivative of f at 0 is 1.

- For $h < 0$, $|h| = -h$, so

$$\begin{aligned}\frac{f(0+h) - f(0)}{h} &= \frac{-h - 0}{h} \\ &= -1.\end{aligned}$$

  The left-hand derivative of f at 0 is -1.

For example, $h = 0.5$ gives $\frac{0.5}{0.5} = 1$, and $h = -0.5$ gives $\frac{0.5}{-0.5} = -1$. The one-sided derivatives differ, since $1 \neq -1$.

Definition 0.13 (Second derivative)**.** Let f be a real-valued function of one variable whose derivative f' (Definition 0.7) exists at every point of an open interval containing c. If f' is differentiable (Definition 0.6) at c, its derivative there, written $f''(c)$ or $\frac{d^2 f}{dx^2}$, is the **second derivative** of f at c.

Example 0.7 (The second derivative of x^3)**.** For $f(x) = x^3$, the power rule (Theorem 0.3) gives $f'(x) = 3x^2$ at every x, and differentiating again, with the constant multiple rule (Theorem 0.2), gives

$$\begin{aligned}f''(x) &= 3 \cdot 2x \\ &= 6x.\end{aligned}$$

So $f''(2) = 12$, $f''(0) = 0$, and $f''(-1) = -6$.

Derivative rules

Theorem 0.1 (Constant rule)**.** *If c is constant with respect to x, then*

$$\frac{\partial}{\partial x} c = 0$$

Video

Jon Krohn's "Calculus for Machine Learning" YouTube playlist has a video on this rule:

- The Derivative of a Constant

Theorem 0.2 (Constant multiple rule)**.** *If a is constant with respect to x and y is a differentiable function of x, then:*

$$\frac{\partial}{\partial x} ay = a\frac{\partial y}{\partial x}$$

Video

Jon Krohn's "Calculus for Machine Learning" YouTube playlist has a video on this rule:

- The Constant Multiple Rule for Derivatives

Theorem 0.3 (Power rule)**.** *For every real number q and every $x > 0$, the derivative of the power x^q is:*

$$\frac{\partial}{\partial x} x^q = qx^{q-1}$$

When q is a positive integer, the same formula holds for every real x.

Videos

Jon Krohn's "Calculus for Machine Learning" YouTube playlist has videos on this rule, on the sum rule (the derivative of a sum is the sum of the derivatives), and on exercises that combine the rules so far:

- The Power Rule for Derivatives
- The Sum Rule for Derivatives
- Exercises on Derivative Rules

Exercise 0.1 (Derivative of polynomial and power functions)**.** Adapted from Miller (2016), Question 1.1.1.
Find the derivative of

$$f(x) = 4x^5 + 3x^2 + x^{1/3}$$

Solution

Solution 0.1. Differentiate term by term using the constant multiple rule (Theorem 0.2) and the power rule (Theorem 0.3):

$$\begin{aligned}
\frac{d}{dx}(4x^5) &= 4 \cdot (5x^4) \\
&= 20x^4 \\
\frac{d}{dx}(3x^2) &= 3 \cdot (2x) \\
&= 6x \\
\frac{d}{dx}(x^{1/3}) &= \frac{1}{3}x^{1/3-1} \\
&= \frac{1}{3}x^{-2/3}
\end{aligned}$$

Combining the three terms yields:

$$f'(x) = 20x^4 + 6x + \frac{1}{3}x^{-2/3}$$

Exercise 0.2 (Finding points with a prescribed slope)**.** Adapted from Miller (2016), Question 1.1.14.
Let $f(x) = \frac{1}{3}x^3 + x^2 - x - 1$. Find all points on the graph of f where the slope is:

1. -1
2. 2
3. 0

Solution

Solution 0.2. The slope is given by the derivative:

$$f'(x) = x^2 + 2x - 1$$

1. Slope -1: Solve

$$\begin{aligned} x^2 + 2x - 1 = -1 &\iff x^2 + 2x = 0 \\ &\iff x(x+2) = 0. \end{aligned}$$

The roots are $x = 0$ and $x = -2$. Evaluating f: $f(0) = -1$ and

$$\begin{aligned} f(-2) &= -\frac{8}{3} + 4 + 2 - 1 \\ &= \frac{7}{3}. \end{aligned}$$

The points are $(0, -1)$ and $(-2, 7/3)$.

2. Slope 2: Solve

$$\begin{aligned} x^2 + 2x - 1 = 2 &\iff x^2 + 2x - 3 = 0 \\ &\iff (x+3)(x-1) = 0. \end{aligned}$$

The roots are $x = -3$ and $x = 1$. Evaluating f:

$$\begin{aligned} f(-3) &= -9 + 9 + 3 - 1 \\ &= 2 \end{aligned}$$

and

$$\begin{aligned} f(1) &= \frac{1}{3} + 1 - 1 - 1 \\ &= -\frac{2}{3}. \end{aligned}$$

The points are $(-3, 2)$ and $(1, -2/3)$.

3. Slope 0: Solve $x^2 + 2x - 1 = 0$. By the quadratic formula:

$$\begin{aligned} x &= \frac{-2 \pm \sqrt{4 - 4(1)(-1)}}{2} \\ &= \frac{-2 \pm \sqrt{8}}{2} \\ &= -1 \pm \sqrt{2} \end{aligned}$$

Theorem 0.4 (Derivative of natural logarithm)**.** *For every $x > 0$, the derivative of the natural logarithm is:*

$$\begin{aligned}\log'\{x\} &= \frac{1}{x} \\ &= x^{-1}\end{aligned}$$

Exercise 0.3 (Derivative after logarithmic simplification)**.** Adapted from Miller (2016), Question 1.1.4.
Find the derivative of

$$f(x) = \log(4x) - \log(2x)$$

for $x > 0$.

Solution

Solution 0.3. Simplify before differentiating using the quotient property of logarithms:

$$\begin{aligned}f(x) &= \log(4x) - \log(2x) \\ &= \log\left(\frac{4x}{2x}\right) \\ &= \log 2\end{aligned}$$

Because $\log 2$ is constant with respect to x, the constant rule (Theorem 0.1) gives:

$$f'(x) = 0$$

Remark: While one could apply the chain rule separately to each term, giving

$$\begin{aligned}\frac{4}{4x} - \frac{2}{2x} &= \frac{1}{x} - \frac{1}{x} \\ &= 0,\end{aligned}$$

simplifying algebraically first is faster and prevents arithmetic errors.

Theorem 0.5 (Derivative of exponential)**.** *For every real x, the derivative of the exponential function is:*

$$\exp'\{x\} = \exp\{x\}$$

Theorem 0.6 (Derivative of sine)**.** *For every real x, the derivative of the sine function is:*

$$\frac{\partial}{\partial x} \sin x = \cos x$$

Theorem 0.7 (Derivative of cosine)**.** *For every real x, the derivative of the cosine function is:*

$$\frac{\partial}{\partial x} \cos x = -\sin x$$

Theorem 0.8 (Product rule)**.** *If a and b are differentiable functions of x, then*

$$(ab)' = ab' + ba'$$

Video

Jon Krohn's "Calculus for Machine Learning" YouTube playlist has a video on this rule:

- The Product Rule for Derivatives

Exercise 0.4 (Derivative of a product)**.** Adapted from Miller (2016), Question 1.1.2. Find the derivative of

$$f(x) = (x^4 + 3x^2 + 8) \cos x$$

Solution

Solution 0.4. Apply the product rule (Theorem 0.8) with:

$$u(x) = x^4 + 3x^2 + 8, \qquad v(x) = \cos x$$

Their derivatives are:

$$u'(x) = 4x^3 + 6x, \qquad v'(x) = -\sin x$$

By the product rule:

$$\begin{aligned} f'(x) &= u'(x)v(x) + u(x)v'(x) \\ &= (4x^3 + 6x) \cos x - (x^4 + 3x^2 + 8) \sin x \end{aligned}$$

Exercise 0.5 (Second derivative of a product)**.** Adapted from Miller (2016), Question 1.1.15.
Find the second derivative of

$$f(x) = (x^4 + 3x^2 + 8)\cos x$$

Solution

Solution 0.5. From Exercise 0.4, the first derivative is:

$$f'(x) = (4x^3 + 6x)\cos x - (x^4 + 3x^2 + 8)\sin x$$

Differentiate each term via the product rule:

$$\frac{d}{dx}\left[(4x^3 + 6x)\cos x\right] = (12x^2 + 6)\cos x - (4x^3 + 6x)\sin x$$
$$\frac{d}{dx}\left[-(x^4 + 3x^2 + 8)\sin x\right] = -(4x^3 + 6x)\sin x - (x^4 + 3x^2 + 8)\cos x$$

Combining and grouping like trigonometric terms:

$$\begin{aligned} f''(x) &= \left[(12x^2 + 6) - (x^4 + 3x^2 + 8)\right]\cos x - 2(4x^3 + 6x)\sin x \\ &= (-x^4 + 9x^2 - 2)\cos x - (8x^3 + 12x)\sin x \end{aligned}$$

Theorem 0.9 (Quotient rule)**.** *If a and b are differentiable functions of x and $b \neq 0$, then*

$$(a/b)' = a'/b - (a/b^2)b'$$

Video

Jon Krohn's "Calculus for Machine Learning" YouTube playlist has a video on this rule:

- The Quotient Rule for Derivatives

Exercise 0.6 (Derivative of a rational function)**.** Adapted from Miller (2016), Question 1.1.12.
Find the derivative of

$$f(x) = \frac{x^2 - 1}{x - 1}$$

for $x \neq 1$.

Solution

Solution 0.6. Factor the numerator before differentiating:

$$\begin{aligned} f(x) &= \frac{(x-1)(x+1)}{x-1} \\ &= x+1 \qquad (x \neq 1) \end{aligned}$$

Differentiating directly gives $f'(x) = 1$.
Alternative (quotient rule): Using Theorem 0.9 with $u(x) = x^2 - 1$ and $v(x) = x - 1$:

$$\begin{aligned} f'(x) &= \frac{2x(x-1) - (x^2-1)(1)}{(x-1)^2} \\ &= \frac{x^2 - 2x + 1}{(x-1)^2} \\ &= \frac{(x-1)^2}{(x-1)^2} \\ &= 1 \end{aligned}$$

Both methods yield 1, but factoring first eliminates tedious algebraic simplification.

Theorem 0.10 (Chain rule)**.** *If a is a differentiable function of b, and b is a differentiable function of c, then a is a differentiable function of c, and*

$$\begin{aligned} \frac{da}{dc} &= \frac{da}{db}\frac{db}{dc} \\ &= \frac{db}{dc}\frac{da}{db} \end{aligned}$$

or in Lagrange's notation (Definition 0.7), if g is differentiable at x and f is differentiable at $g(x)$, then for the composition $f \circ g$, with inner function g and outer function f:

$$(f(g(x)))' = g'(x)f'(g(x))$$

Exercise 0.7 (Derivative of the standard Gaussian kernel)**.** Adapted from Miller (2016), Question 1.1.5.
Find the derivative of

$$f(x) = \exp\left\{-\frac{x^2}{2}\right\}$$

Solution

Solution 0.7. Apply the chain rule (Theorem 0.10) with outer function $\exp\{u\}$ and inner function $u(x) = -x^2/2$. Since $\frac{d}{du}\exp\{u\} = \exp\{u\}$ (Theorem 0.5) and $u'(x) = -x$:

$$\begin{aligned} f'(x) &= u'(x)\exp\{u(x)\} \\ &= -x\exp\left\{-\frac{x^2}{2}\right\} \end{aligned}$$

Theorem 0.11 (Derivative of the standard Gaussian kernel)**.** *For* $f(x) = \exp\left\{-\frac{x^2}{2}\right\}$ *and every* $x \in \mathbb{R}$*,*

$$f'(x) = -x\exp\left\{-\frac{x^2}{2}\right\}$$

Proof

Proof. This is the solution to Exercise 0.7, worked in Solution 0.7. □

Exercise 0.8 (Second derivative of the Gaussian kernel)**.** Adapted from Miller (2016), Question 1.1.6.
Find the second derivative of

$$f(x) = \exp\left\{-\frac{x^2}{2}\right\}$$

Solution

Solution 0.8. From Theorem 0.11, the first derivative is $f'(x) = -x\exp\{-x^2/2\}$. Apply the product rule (Theorem 0.8) to $u(x) = -x$ and $v(x) = \exp\{-x^2/2\}$:

$$\begin{aligned} u'(x) &= -1 \\ v'(x) &= -x\exp\left\{-\frac{x^2}{2}\right\} \end{aligned}$$

Therefore:

$$
\begin{aligned}
f''(x) &= u'(x)v(x) + u(x)v'(x) \\
&= (-1)\exp\left\{-\frac{x^2}{2}\right\} + (-x)\left[-x\exp\left\{-\frac{x^2}{2}\right\}\right] \\
&= (x^2-1)\exp\left\{-\frac{x^2}{2}\right\}
\end{aligned}
$$

Theorem 0.12 (Second derivative of the standard Gaussian kernel)**.** *For* $f(x) = \exp\left\{-\frac{x^2}{2}\right\}$ *and every* $x \in \mathbb{R}$,

$$
f''(x) = (x^2-1)\exp\left\{-\frac{x^2}{2}\right\}
$$

Proof

Proof. This is the solution to Exercise 0.8, worked in Solution 0.8. □

Exercise 0.9 (Combining product and chain rules)**.** Adapted from Miller (2016), Question 1.1.7.
Find the derivative of

$$
f(x) = \exp\{x^8\}\cos(3x^4)
$$

Solution

Solution 0.9. Apply the product rule (Theorem 0.8) to $u(x) = \exp\{x^8\}$ and $v(x) = \cos(3x^4)$:

$$
f'(x) = u'(x)v(x) + u(x)v'(x)
$$

Compute $u'(x)$ and $v'(x)$ via the chain rule (Theorem 0.10):

$$
\begin{aligned}
u'(x) &= 8x^7\exp\{x^8\} \\
v'(x) &= -12x^3\sin(3x^4)
\end{aligned}
$$

Substitute both derivatives into the product rule:

$$
\begin{aligned}
f'(x) &= 8x^7\exp\{x^8\}\cos(3x^4) - 12x^3\exp\{x^8\}\sin(3x^4) \\
&= 4x^3\exp\{x^8\}[2x^4\cos(3x^4) - 3\sin(3x^4)]
\end{aligned}
$$

Exercise 0.10 (Generalized power rule)**.** Adapted from Miller (2016), Question 1.1.13. Find the derivative of

$$\begin{aligned} f(x) &= \sqrt[3]{(5x-2)^2} \\ &= (5x-2)^{2/3} \end{aligned}$$

for $x \neq 2/5$.

Solution

Solution 0.10. Apply the generalized power rule $\frac{d}{dx}[g(x)]^r = r[g(x)]^{r-1}g'(x)$ with $g(x) = 5x - 2$ and $r = 2/3$:

$$\begin{aligned} f'(x) &= \frac{2}{3}(5x-2)^{2/3-1} \cdot 5 \\ &= \frac{10}{3}(5x-2)^{-1/3} \\ &= \frac{10}{3(5x-2)^{1/3}} \end{aligned}$$

Remark: Omitting the inner derivative $g'(x) = 5$ is a frequent mistake when applying the generalized power rule.

Corollary 0.1 (Chain rule for logarithms)**.** *If f is differentiable at x and $f(x) > 0$, then*

$$\frac{d}{dx}\log\{f(x)\} = \frac{f'(x)}{f(x)}$$

Proof

Proof. Apply Theorem 0.10 and Theorem 0.4:

$$\begin{aligned} \frac{d}{dx}\log\{f(x)\} &= f'(x) \cdot \log'\{f(x)\} && \text{(chain rule, with } g = f \text{ and outer function } \log\text{)} \\ &= f'(x) \cdot \frac{1}{f(x)} && \text{(derivative of } \log\text{, valid because } f(x) > 0\text{)} \\ &= \frac{f'(x)}{f(x)} && \text{(multiply)} \end{aligned}$$

□

Videos

Jon Krohn's "Calculus for Machine Learning" YouTube playlist has videos on this rule and on exercises that combine it with the other rules:

- The Chain Rule for Derivatives
- The Power Rule on a Function Chain
- Advanced Exercises on Derivative Rules

Exercise 0.11 (Chain rule with logarithm)**.** Adapted from Miller (2016), Question 1.1.3. Find the derivative of

$$f(x) = \log(1 - x^2)$$

for $x \in (-1, 1)$.

Solution

Solution 0.11. Apply the chain rule for logarithms (Corollary 0.1): if $g(x) = 1 - x^2$, then $g'(x) = -2x$, and

$$\begin{aligned} f'(x) &= \frac{g'(x)}{g(x)} \\ &= -\frac{2x}{1 - x^2} \end{aligned}$$

Remark: A common pitfall in chain rule problems is evaluating the outer derivative at x rather than at the inner value $g(x)$. The denominator is $g(x) = 1 - x^2$, not x.

Exercise 0.12 (Second derivative with an added constant)**.** Adapted from Miller (2016), Question 1.1.9.
Find the second derivative of

$$f(x) = \log x + \sqrt{162}$$

for $x > 0$.

Solution

Solution 0.12. Because $\sqrt{162}$ is constant with respect to x, its derivative is zero (Theorem 0.1):

$$\begin{aligned} f'(x) &= \frac{1}{x} \\ &= x^{-1} \end{aligned}$$

Differentiating again via the power rule (Theorem 0.3):

$$\begin{aligned} f''(x) &= -x^{-2} \\ &= -\frac{1}{x^2} \end{aligned}$$

Studying a new function

Definition 0.14 (Logarithmic derivative)**.** The **logarithmic derivative** $\frac{f'(x)}{f(x)}$ of a function f of a real variable equals $\frac{d}{dx}\log\{f(x)\}$ by Corollary 0.1 and measures the rate of change of f relative to its size; it needs $f(x) > 0$, and for $f(x) < 0$ the same expression is the derivative of $\log\{|f(x)|\}$.

Definition 0.15 (Profile of a function)**.** The **profile** of a function f of a real variable is the list of related functions and values to work out when meeting f for the first time:

- the logarithm $\log\{f(x)\}$, where $f(x) > 0$;
- the derivatives $f'(x)$, $f''(x)$, and so on, as far as they are informative;
- the logarithmic derivative $\frac{f'(x)}{f(x)}$;
- the inverse function f^{-1} (definition), which exists on the range of f when f is one-to-one, and otherwise only after restricting the domain of f;
- the complement $1 - f(x)$, which is informative when f takes values in $[0, 1]$, as a probability does;
- the reciprocal $\frac{1}{f(x)}$, where $f(x) \neq 0$ (not to be confused with f^{-1});
- an antiderivative of f, a function F with $F' = f$ (integration);
- the values $f(0)$ and $f(1)$, and the limits of $f(x)$ as $x \to \infty$ and as $x \to -\infty$, at each of these that lies in the domain of f or exists;
- the graph of f.

Example 0.8 (The profile of the logistic function)**.** The **logistic function** (also called expit or the sigmoid) is

$$f(x) = \frac{1}{1 + e^{-x}}.$$

The profile in Definition 0.15 of f is as follows. Write $E = e^{-x}$, so that $f = \frac{1}{1+E}$ and $1 - f = \frac{E}{1+E}$.

- **Logarithm:** $\log\{f(x)\} = -\log\{1 + e^{-x}\}$, defined for every real x because $f(x) > 0$.
- **Derivative:** by the chain rule,

$$\begin{aligned} f'(x) &= \frac{e^{-x}}{(1+e^{-x})^2} && \text{(chain rule applied to } (1+e^{-x})^{-1}) \\ &= \frac{1}{1+E} \cdot \frac{E}{1+E} && \text{(split the square)} \\ &= f(x)\,(1 - f(x)). && \text{(identify } f \text{ and } 1-f) \end{aligned}$$

  Differentiating again with the product rule gives $f''(x) = f'(1-f) - ff' = f(x)(1-f(x))(1-2f(x))$.
- **Logarithmic derivative:** $\frac{f'(x)}{f(x)} = 1 - f(x)$, so the relative rate of change of f is the complement of f itself.
- **Inverse:** for $y \in (0,1)$, solving $y = \frac{1}{1+e^{-x}}$ for x gives

$$\begin{aligned} \frac{1}{y} &= 1 + e^{-x} \\ e^{-x} &= \frac{1-y}{y} \\ x &= \log\left\{\frac{y}{1-y}\right\}. \end{aligned}$$

  So $f^{-1}(y) = \text{logit}\{y\}$, the log-odds. The inverse exists because $f' = f(1-f) > 0$, so f is strictly increasing and therefore one-to-one.
- **Complement:** $1 - f(x) = \frac{e^{-x}}{1+e^{-x}} = \frac{1}{e^{x}+1} = f(-x)$.
- **Reciprocal:** $\frac{1}{f(x)} = 1 + e^{-x}$.
- **Antiderivative:** $F(x) = \log\{1 + e^{x}\}$ satisfies $F'(x) = \frac{e^{x}}{1+e^{x}} = f(x)$, so $\int f(x)\,dx = \log\{1 + e^{x}\} + C$.
- **Key values:** $f(0) = \frac{1}{1+1} = \frac{1}{2}$ and $f(1) = \frac{1}{1+e^{-1}}$. As $x \to \infty$, $e^{-x} \to 0$, so $f(x) \to 1$; as $x \to -\infty$, $e^{-x} \to \infty$, so $f(x) \to 0$.

The code checks these results numerically, using the base R functions `plogis()` for f and `qlogis()` for f^{-1}, and `D()` to differentiate symbolically.

```
f <- plogis
x <- c(-2, -0.5, 0, 1, 3)

f_expr <- quote(1 / (1 + exp(-x)))
f_prime <- eval(D(f_expr, "x"), list(x = x))
f_second <- eval(D(D(f_expr, "x"), "x"), list(x = x))

checks <- c(
  derivative = max(abs(f_prime - f(x) * (1 - f(x)))),
  second_derivative = max(abs(f_second - f(x) * (1 - f(x)) * (1 - 2 * f(x)))),
  log_derivative = max(abs(f_prime / f(x) - (1 - f(x)))),
  inverse = max(abs(qlogis(f(x)) - x)),
  complement = max(abs(1 - f(x) - f(-x))),
  reciprocal = max(abs(1 / f(x) - (1 + exp(-x)))),
  antiderivative = abs(
    integrate(f, 0, 1)$value - (log(1 + exp(1)) - log(2))
  )
)
signif(checks, 3)
#>        derivative second_derivative    log_derivative           inverse
#>          1.46e-16          1.18e-16          1.53e-16          3.11e-15
#>        complement        reciprocal    antiderivative
#>          1.46e-16          0.00e+00          0.00e+00
stopifnot(all(checks < 1e-14))

key_values <- c(f(0), f(1), f(10), f(-10))
```

Each entry of `checks` is the largest discrepancy between a formula above and a numerical computation; every one is below 10^{-14} (the chunk stops with an error otherwise), which is rounding error. The key values are $f(0) = 0.5$ and $f(1) \approx 0.7311$. At $x = 10$ and $x = -10$ the function is already close to its limits, $f(10) \approx 0.99995$ and $f(-10) \approx 0.0000454$. Figure 1 graphs f, f' and $\log\{f\}$.

```r
grid <- seq(-6, 6, length.out = 241)
panels <- data.frame(
  x = rep(grid, 3),
  panel = factor(
    rep(c("f(x)", "f'(x)", "log f(x)"), each = length(grid)),
    levels = c("f(x)", "f'(x)", "log f(x)")
  ),
  value = c(f(grid), f(grid) * (1 - f(grid)), log(f(grid)))
)
ggplot2::ggplot(panels) +
  ggplot2::aes(x = x, y = value) +
  ggplot2::geom_line() +
  ggplot2::facet_wrap(~panel, scales = "free_y") +
  ggplot2::labs(x = "x", y = NULL)
```

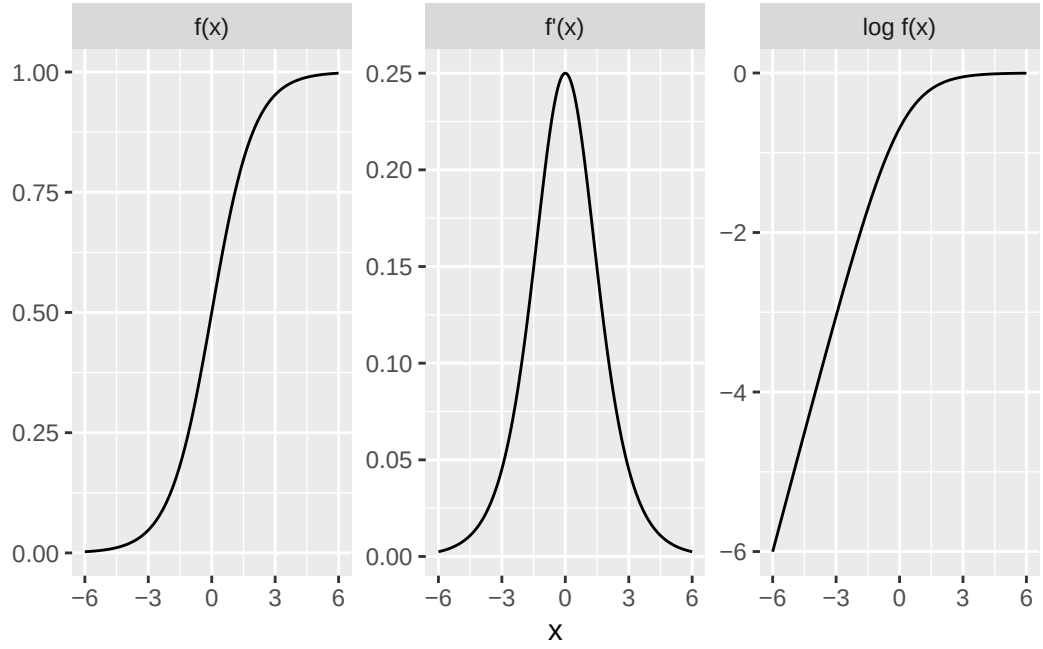


Figure 1: The logistic function, its derivative and its logarithm

Tangent lines

Definition 0.16 (Tangent line)**.** Let f be differentiable at c (Definition 0.6). The **tangent line** to the graph of f at c is the graph of the affine function

$$x \mapsto f(c) + f'(c)\,(x - c),$$

the line through the point $(c, f(c))$ with slope $f'(c)$. The slope $f'(c)$ is also called the **tangent slope** of f at c.

Example 0.9 (The tangent line to x^2 at 1)**.** For $f(x) = x^2$ at $c = 1$, $f(1) = 1$ and $f'(1) = 2$ (Example 0.5), so the tangent line is the graph of

$$\begin{aligned} x &\mapsto 1 + 2\,(x - 1) && \text{(substitute } f(1) = 1 \text{ and } f'(1) = 2\text{)} \\ &= 2x - 1 && \text{(distribute, and } 1 - 2 = -1\text{)} \end{aligned}$$

At $x = 1.1$ the tangent line has height $2(1.1) - 1 = 1.2$, close to the curve's height $f(1.1) = 1.21$; the gap, $1.21 - 1.2 = 0.01$, is $(1.1 - 1)^2$.

Linear approximation

For a differentiable function f and a small step ε,

$$f(w + \varepsilon) \approx f(w) + \varepsilon \frac{d}{dw} f(w) \tag{1}$$

Definition 0.17 (Linear approximation (first-order Taylor approximation))**.** For a function f that is differentiable at w, the **linear approximation** of f at w (also called the **first-order Taylor approximation**) is the function of the step ε

$$\hat{f}_w(\varepsilon) = f(w) + \varepsilon \frac{d}{dw} f(w)$$

so Equation 1 says $f(w + \varepsilon) \approx \hat{f}_w(\varepsilon)$ for small ε. Figure 2 illustrates this approximation interactively for a quadratic function.

ℹ Remark (The "linear" approximation is affine)

Remark 0.2 (The "linear" approximation is affine). The linear approximation $\hat{f}_w$ (Definition 0.17) is an affine function of the step ε: its slope is $\frac{d}{dw} f(w)$ and its intercept is $f(w)$.

It is linear in ε only when $f(w) = 0$. The name "linear approximation" uses "linear" in the looser sense of elementary algebra (remark). Boyd and Vandenberghe (2018), section 2.2, treats the first-order Taylor approximation as an affine function for functions of several variables as well.

(This plot is interactive; see the HTML or slide version of this page to view it.)

Figure 2: The linear approximation Equation 1 for $f(w) = w^2 - 4w + 7$, at any w and step ε.

Exercise 0.13 (Tangent line approximation)**.** Adapted from Miller (2016), Question 1.1.8.
Let $f(x) = 4x + \sqrt{2}\cos x$.

1. Compute $f'(x)$.
2. Find the equation of the tangent line to the curve $y = f(x)$ at $x = \pi/4$.
3. Use the tangent line to approximate $f(\pi/4 + 0.01)$, and compare that approximation with the exact value.

Solution

Solution 0.13. **1.** Differentiate term by term:

$$f'(x) = 4 - \sqrt{2}\sin x$$

2. Evaluate f and f' at $x_0 = \pi/4$:

$$\begin{aligned} f(\pi/4) &= 4\left(\frac{\pi}{4}\right) + \sqrt{2}\cos\left(\frac{\pi}{4}\right) \\ &= \pi + \sqrt{2}\left(\frac{\sqrt{2}}{2}\right) \\ &= \pi + 1 \\ f'(\pi/4) &= 4 - \sqrt{2}\sin\left(\frac{\pi}{4}\right) \\ &= 4 - \sqrt{2}\left(\frac{\sqrt{2}}{2}\right) \\ &= 4 - 1 \\ &= 3 \end{aligned}$$

Using the point-slope formula, the tangent line at $(\pi/4, \pi + 1)$ is:

$$y - (\pi + 1) = 3\left(x - \frac{\pi}{4}\right) \implies y = \pi + 1 + 3\left(x - \frac{\pi}{4}\right)$$

3. At $x = \pi/4 + 0.01$, the linear approximation (Equation 1) gives:

$$f(\pi/4 + 0.01) \approx (\pi + 1) + 3(0.01) = \pi + 1.03 \approx 4.171593$$

The exact value is:

$$f(\pi/4 + 0.01) = 4\left(\frac{\pi}{4} + 0.01\right) + \sqrt{2}\cos\left(\frac{\pi}{4} + 0.01\right) \approx 4.171543$$

The approximation error is $|4.171593 - 4.171543| \approx 0.00005$, on the order of $(0.01)^2 = 10^{-4}$.

Critical points and optimization

Definition 0.18 (Flat point (stationary point))**.** Let f be differentiable at c (Definition 0.6). If $f'(c) = 0$, then c is a **flat point** of f (also called a **stationary point**).

Example 0.10 (Flat points, and a point that is not one)**.**

- $f(w) = w^3$ has $f'(w) = 3w^2$, which is 0 at $w = 0$, so 0 is a flat point. Yet $f(0) = 0$ is neither the minimum nor the maximum of the values f takes on any open interval around 0: $f(w) < 0$ for $w < 0$ and $f(w) > 0$ for $w > 0$.
- $h(w) = w^2$ has $h'(1) = 2 \neq 0$, so 1 is not a flat point.

Exercise 0.14 (Find the flat point, and check the approximation)**.** Let $f(w) = w^2 - 4w + 7$.

1. Differentiate f.
2. Find the flat point w of f, and say whether $f(w)$ is the minimum or the maximum of the values of f.
3. Evaluate the derivative at $w = 1$, use Equation 1 to predict $f(1.01)$, and compare that prediction with the exact value.

Solution

Solution 0.14. **1.** Term by term:

$$\frac{df}{dw} = 2w - 4$$

2. Set it to zero: $2w - 4 = 0$ gives $w = 2$.

$$\begin{aligned} f(2) &= 4 - 8 + 7 \\ &= 3 \end{aligned}$$

is the minimum of the values of f. Completing the square, $f(w) = (w-2)^2 + 3$, since

$$\begin{aligned} (w-2)^2 + 3 &= w^2 - 4w + 4 + 3 \\ &= w^2 - 4w + 7, \end{aligned}$$

and $(w-2)^2 \geq 0$, so $f(w) \geq 3 = f(2)$ for every w. Equivalently, the derivative is negative below $w = 2$ and positive above it, so the function falls into that point and rises out of it.
3. At $w = 1$ the derivative is $2(1) - 4 = -2$, so f is falling there. With $\varepsilon = 0.01$, Equation 1 predicts a change of $(0.01)(-2) = -0.02$, from

$$\begin{aligned} f(1) &= 1 - 4 + 7 \\ &= 4 \end{aligned}$$

to 3.98. The exact value is

$$\begin{aligned} f(1.01) &= (1.01)^2 - 4(1.01) + 7 \\ &= 1.0201 - 4.04 + 7 \\ &= 3.9801 \end{aligned}$$

a change of -0.0199. The prediction is off by 0.0001, which is ε^2: the linear approximation drops everything of that order and smaller, so halving the step quarters the error. That trade is the whole bargain of gradient descent, the step-by-step method of fitting models defined on the optimization page. We take a step in the direction the derivative recommends, and the recommendation is trustworthy only as far as the step is small.

Definition 0.19 (Critical point)**.** Let f be a function defined on an open interval containing c. The point c is a **critical point** of f if either $f'(c) = 0$ or f is not differentiable at c (Definition 0.6).

Example 0.11 (Critical points that are and are not flat points)**.**

- $g(w) = |w|$ has no derivative at $w = 0$: its one-sided derivatives there are 1 and -1 (Example 0.6), so the difference quotient $\frac{|h|-0}{h}$ has no limit as $h \to 0$ (Definition 0.4). So 0 is a critical point of g but not a flat point.
- $f(w) = w^3$ has

$$\begin{aligned} f'(0) &= 3 \cdot 0^2 \\ &= 0, \end{aligned}$$

so 0 is a flat point of f, and hence a critical point.

- $h(w) = w^2$ is differentiable everywhere, with $h'(w) = 2w$, which is 0 only at $w = 0$. So 0 is the only critical point of h; for example, $h'(1) = 2 \neq 0$, so 1 is not one.

Theorem 0.13 (Every flat point is a critical point)**.** *If c is a flat point of f (Definition 0.18), then c is a critical point of f (Definition 0.19).*

Proof

Proof. A flat point c has f differentiable at c and $f'(c) = 0$. The condition "$f'(c) = 0$" is the first alternative in Definition 0.19, so c is a critical point. □

Example 0.12 (The flat points of a cubic, and a critical point that is not flat)**.** For $f(w) = w^3 - 3w$, the derivative is $f'(w) = 3w^2 - 3$. The code differentiates f with `D()` and evaluates f' at several points. For $g(w) = |w|$, it computes the difference quotients at 0 from both sides.

```
f_prime <- D(quote(w^3 - 3 * w), "w")
points <- c(-2, -1, 0, 1, 2)
derivs <- data.frame(w = points, f_prime = eval(f_prime, list(w = points)))
derivs
#>    w f_prime
#> 1 -2       9
#> 2 -1       0
#> 3  0      -3
#> 4  1       0
#> 5  2       9
flat <- derivs$w[derivs$f_prime == 0]

h <- c(-1e-3, 1e-3)
quotients <- abs(h) / h
quotients
#> [1] -1  1
```

The derivative is 0 at 2 of the 5 points, namely $w = -1$ and $w = 1$. These are flat points of f, so by Theorem 0.13 they are critical points. The difference quotients of g at 0 are -1 from the left and 1 from the right, so g has no derivative at 0. Thus 0 is a critical point of g (Definition 0.19), but not a flat point.

Exercise 0.15 (Maximizing a gamma-family kernel)**.** Adapted from Miller (2016), Question 1.1.10.
Find the maximum value of

$$f(x) = x^4 e^{-x}$$

on the interval $[0, \infty)$.

Solution

Solution 0.15. Differentiate using the product rule:

$$\begin{aligned} f'(x) &= 4x^3 e^{-x} - x^4 e^{-x} \\ &= x^3 e^{-x}(4 - x) \end{aligned}$$

On $[0, \infty)$, $e^{-x} > 0$, so $f'(x) = 0$ only at $x = 0$ and $x = 4$.
Evaluate f at the boundary and critical points:

- At $x = 0$: $f(0) = 0$.
- At $x = 4$:

$$\begin{aligned} f(4) &= 4^4 e^{-4} \\ &= 256 e^{-4} \\ &\approx 4.6888. \end{aligned}$$

- As $x \to \infty$: exponential decay dominates polynomial growth, so $\lim_{x \to \infty} x^4 e^{-x} = 0$.

Since $f(x) \geq 0$ for all $x \geq 0$ and $f(4) > 0$, the global maximum occurs at $x = 4$, with maximum value $256/e^4$.

Exercise 0.16 (Classifying critical points with the second derivative test)**.** Adapted from Miller (2016), Question 1.1.11.
Find the critical points of

$$f(x) = 4x^3 - 3x^2$$

and decide whether each is a local maximum, a local minimum, or an inflection point.

Solution

Solution 0.16. Compute the first derivative:

$$\begin{aligned} f'(x) &= 12x^2 - 6x \\ &= 6x(2x-1) \end{aligned}$$

Setting $f'(x) = 0$ gives critical points at $x = 0$ and $x = 1/2$.
Compute the second derivative:

$$f''(x) = 24x - 6$$

Apply the second derivative test at each critical point:

- At $x = 0$: $f''(0) = -6 < 0$, so $x = 0$ is a strict local maximum.
- At $x = 1/2$:

$$\begin{aligned} f''(1/2) &= 24(1/2) - 6 \\ &= 6 \\ &> 0, \end{aligned}$$

  so $x = 1/2$ is a strict local minimum.

Inflection points occur where the second derivative changes sign ($f''(x) = 0$ at $x = 1/4$). Because f'' is non-zero at both critical points, neither is an inflection point.

Taylor series

Definition 0.20 (Taylor polynomial)**.** Let f be a function that has at least n derivatives at a point x_0. The **Taylor polynomial** of degree n for f centered at x_0 is:

$$\begin{aligned} P_n(x) &= \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k \\ &= f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2!}(x - x_0)^2 + \cdots + \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n \end{aligned} \tag{2}$$

Example 0.13 (Taylor series of the exponential function)**.** For $f(x) = e^x$ centered at $x_0 = 0$, every derivative is

$$\begin{aligned} f^{(k)}(0) &= e^0 \\ &= 1. \end{aligned}$$

The degree-n Maclaurin polynomial is:

$$\begin{aligned} P_n(x) &= 1 + x + \frac{x^2}{2!} + \cdots + \frac{x^n}{n!} \\ &= \sum_{k=0}^{n} \frac{x^k}{k!} \end{aligned}$$

Because the remainder $R_n(x) \to 0$ as $n \to \infty$ for all $x \in \mathbb{R}$, the Taylor series converges everywhere to e^x:

$$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$$

Definition 0.21 (Maclaurin polynomial)**.** When $x_0 = 0$, the Taylor polynomial $P_n(x)$ (Definition 0.20) is called the **Maclaurin polynomial**.

Definition 0.22 (Taylor series)**.** Let f and x_0 be as in Definition 0.20. When f is infinitely differentiable and the series converges to $f(x)$ on an open interval containing x_0, the infinite sum $\sum_{k=0}^{\infty} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k$ is the **Taylor series** of f centered at x_0.

Video

Chapter 11 of the *Essence of calculus* series by 3Blue1Brown develops the geometric and polynomial intuition behind Taylor approximations:

- Taylor series shows how higher-order derivatives match the curvature and rate of change of a function near a point to build polynomial approximations (Definition 0.20).

Exercise 0.17 (Taylor polynomial of a polynomial at the origin)**.** Adapted from Miller (2016), Question 1.1.16.
Find the first five terms (up to degree 4) of the Taylor series for

$$f(x) = x^8 + x^4 + 3$$

at $x_0 = 0$.

Solution

Solution 0.17. Evaluate f and its first four derivatives at $x = 0$:

$$\begin{aligned} f(x) &= x^8 + x^4 + 3 && \implies f(0) = 3 \\ f'(x) &= 8x^7 + 4x^3 && \implies f'(0) = 0 \\ f''(x) &= 56x^6 + 12x^2 && \implies f''(0) = 0 \\ f'''(x) &= 336x^5 + 24x && \implies f'''(0) = 0 \\ f^{(4)}(x) &= 1680x^4 + 24 && \implies f^{(4)}(0) = 24 \end{aligned}$$

By Equation 2:

$$\begin{aligned} P_4(x) &= 3 + 0x + \frac{0}{2!}x^2 + \frac{0}{3!}x^3 + \frac{24}{4!}x^4 \\ &= 3 + x^4 \end{aligned}$$

Because f is already a polynomial centered at 0, its Taylor polynomial of degree 4 simply collects its terms of degree at most 4.

Exercise 0.18 (Taylor expansion of a polynomial at a nonzero point)**.** Adapted from Miller (2016), Question 1.1.17.
Find the first three terms (up to degree 2) of the Taylor series for

$$f(x) = x^8 + x^4 + 3$$

at $x_0 = 1$.

Solution

Solution 0.18. Evaluate f and its first two derivatives at $x_0 = 1$:

$$\begin{aligned} f(1) &= 1^8 + 1^4 + 3 \\ &= 5 \\ f'(1) &= 8(1)^7 + 4(1)^3 \\ &= 12 \\ f''(1) &= 56(1)^6 + 12(1)^2 \\ &= 68 \end{aligned}$$

The degree-2 Taylor polynomial is:

$$\begin{aligned} P_2(x) &= f(1) + f'(1)(x-1) + \frac{f''(1)}{2!}(x-1)^2 \\ &= 5 + 12(x-1) + 34(x-1)^2 \end{aligned}$$

Example 0.14 (Taylor series of the cosine function)**.** For $f(x) = \cos x$ centered at $x_0 = 0$, the derivatives follow a repeating cycle of length 4:

$$\begin{aligned} f(0) &= \cos(0) \\ &= 1, \\ f'(0) &= -\sin(0) \\ &= 0, \\ f''(0) &= -\cos(0) \\ &= -1, \\ f'''(0) &= \sin(0) \\ &= 0, \end{aligned}$$

and in general $f^{(2k)}(0) = (-1)^k$ and $f^{(2k+1)}(0) = 0$ for all $k \geq 0$. The Taylor series (Maclaurin series) converges everywhere to $\cos x$:

$$\begin{aligned} \cos x &= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + ... \\ &= \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!} x^{2k} \end{aligned}$$

Example 0.15 (Taylor series of the sine function)**.** For $f(x) = \sin x$ centered at $x_0 = 0$, the derivatives evaluate at 0 to $f^{(2k)}(0) = 0$ and $f^{(2k+1)}(0) = (-1)^k$ for all $k \geq 0$. The Taylor series converges everywhere to $\sin x$:

$$\begin{aligned} \sin x &= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + ... \\ &= \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} x^{2k+1} \end{aligned}$$

Exercise 0.19 (Taylor expansion of a scaled cosine)**.** Adapted from Miller (2016), Question 1.1.18.
Find the terms up to degree 4 in the Taylor series for

$$f(x) = \cos(5x)$$

at $x_0 = 0$.

Solution

Solution 0.19. Recall the standard Maclaurin series for cosine:

$$\cos u = 1 - \frac{u^2}{2!} + \frac{u^4}{4!} - ...$$

Substitute $u = 5x$:

$$\begin{aligned}\cos(5x) &= 1 - \frac{(5x)^2}{2!} + \frac{(5x)^4}{4!} - ... \\ &= 1 - \frac{25}{2}x^2 + \frac{625}{24}x^4 - ...\end{aligned}$$

Remark: Substituting into known Taylor expansions avoids computing high-order derivatives by hand.

Exercise 0.20 (Taylor series of a power of a function)**.** Adapted from Miller (2016), Question 1.1.19.
Find the terms up to degree 4 in the Taylor series for

$$f(x) = \cos^3(5x)$$

at $x_0 = 0$.

Solution

Solution 0.20. From Exercise 0.19, the expansion of $\cos(5x)$ up to order x^4 is:

$$\cos(5x) = 1 - \frac{25}{2}x^2 + \frac{625}{24}x^4 + O(x^6)$$

Cube this expansion using $(1-y)^3 = 1 - 3y + 3y^2 - y^3$ with $y = \frac{25}{2}x^2 - \frac{625}{24}x^4$:

$$\begin{aligned}\cos^3(5x) &= 1 - 3\left(\frac{25}{2}x^2 - \frac{625}{24}x^4\right) + 3\left(\frac{25}{2}x^2\right)^2 + O(x^6) \\ &= 1 - \frac{75}{2}x^2 + \frac{625}{8}x^4 + \frac{1875}{4}x^4 + O(x^6) \\ &= 1 - \frac{75}{2}x^2 + \frac{4375}{8}x^4 + O(x^6)\end{aligned}$$

Exercise 0.21 (Linear Taylor polynomial of the exponential)**.** Adapted from Miller (2016), Question 1.1.20.
Find the first two terms (degree 1) of the Taylor series for

$$f(x) = e^x$$

at $x_0 = 0$.

Solution

Solution 0.21. Since

$$\begin{aligned} f(0) &= e^0 \\ &= 1 \end{aligned}$$

and

$$\begin{aligned} f'(0) &= e^0 \\ &= 1 : \end{aligned}$$

$$\begin{aligned} P_1(x) &= f(0) + f'(0)x \\ &= 1 + x \end{aligned}$$

The full Taylor series is

$$\begin{aligned} e^x &= \sum_{n=0}^{\infty} \frac{x^n}{n!} \\ &= 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \dots . \end{aligned}$$

Exercise 0.22 (Taylor series of a composite exponential)**.** Adapted from Miller (2016), Question 1.1.21.
Find the first six terms (up to degree 5) of the Taylor series for

$$f(x) = \exp\{x^8\}$$

at $x_0 = 0$.

Solution

Solution 0.22. Substitute $u = x^8$ into $e^u = 1 + u + \frac{u^2}{2!} + \dots$:

$$\exp\{x^8\} = 1 + x^8 + \frac{x^{16}}{2} + ...$$

Because the lowest-order variable term is x^8, all coefficients for powers x^1, x^2, x^3, x^4, x^5 are zero:

$$\begin{aligned} P_5(x) &= 1 + 0x + 0x^2 + 0x^3 + 0x^4 + 0x^5 \\ &= 1 \end{aligned}$$

Exercise 0.23 (Taylor expansion of the normal density)**.** Adapted from Miller (2016), Question 1.1.22.
Find the terms up to degree 3 of the Taylor series for the standard normal density

$$\phi(x) = \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{x^2}{2}\right\}$$

at $x_0 = 0$.

Solution

Solution 0.23. Using $e^u = 1 + u + O(u^2)$ with $u = -x^2/2$:

$$\exp\left\{-\frac{x^2}{2}\right\} = 1 - \frac{x^2}{2} + O(x^4)$$

Multiplying by the normalization constant $1/\sqrt{2\pi}$:

$$\phi(x) = \frac{1}{\sqrt{2\pi}} - \frac{x^2}{2\sqrt{2\pi}} + O(x^4)$$

Up to degree 3, the odd powers vanish by symmetry, so:

$$P_3(x) = \frac{1}{\sqrt{2\pi}} - \frac{x^2}{2\sqrt{2\pi}}$$

Theorem 0.14 (Maclaurin polynomial of the standard normal density)**.** *The Maclaurin polynomial (Definition 0.21) of degree 3 of the standard normal density* $\phi(x) = \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{x^2}{2}\right\}$ *is*

$$P_3(x) = \frac{1}{\sqrt{2\pi}} - \frac{x^2}{2\sqrt{2\pi}}$$

Proof

Proof. This is the solution to Exercise 0.23, worked in Solution 0.23. □

Exercise 0.24 (Taylor expansion of the square root)**.** Adapted from Miller (2016), Question 1.1.23.
Find the first three terms (up to degree 2) of the Taylor series for

$$f(x) = \sqrt{x}$$

at $x_0 = 1/3$.

Solution

Solution 0.24. Compute the values at $x_0 = 1/3$:

$$\begin{aligned} f(1/3) &= \left(\frac{1}{3}\right)^{1/2} \\ &= \frac{1}{\sqrt{3}} \\ &= \frac{\sqrt{3}}{3} \\ f'(x) &= \frac{1}{2}x^{-1/2} \\ \implies f'(1/3) &= \frac{1}{2}\sqrt{3} \\ &= \frac{\sqrt{3}}{2} \\ f''(x) &= -\frac{1}{4}x^{-3/2} \\ \implies f''(1/3) &= -\frac{1}{4}\cdot 3\sqrt{3} \\ &= -\frac{3\sqrt{3}}{4} \end{aligned}$$

The degree-2 Taylor polynomial is:

$$P_2(x) = \frac{\sqrt{3}}{3} + \frac{\sqrt{3}}{2}\left(x - \frac{1}{3}\right) - \frac{3\sqrt{3}}{8}\left(x - \frac{1}{3}\right)^2$$

Exercise 0.25 (Taylor expansion of a fractional power)**.** Adapted from Miller (2016), Question 1.1.24.
Find the first three terms of the Taylor series for

$$f(x) = (1+x)^{1/3}$$

at $x_0 = 1/2$.

Solution

Solution 0.25. At $x_0 = 1/2$, $1 + x_0 = 3/2$:

$$\begin{aligned} f(1/2) &= \left(\frac{3}{2}\right)^{1/3} \\ f'(x) &= \frac{1}{3}(1+x)^{-2/3} \\ \implies f'(1/2) &= \frac{1}{3}\left(\frac{3}{2}\right)^{-2/3} \\ &= \frac{1}{3}\left(\frac{2}{3}\right)^{2/3} \\ f''(x) &= -\frac{2}{9}(1+x)^{-5/3} \\ \implies f''(1/2) &= -\frac{2}{9}\left(\frac{3}{2}\right)^{-5/3} \\ &= -\frac{2}{9}\left(\frac{2}{3}\right)^{5/3} \end{aligned}$$

Dividing $f''(1/2)$ by 2! gives $-\frac{1}{9}\left(\frac{2}{3}\right)^{5/3}$. The Taylor polynomial is:

$$P_2(x) = \left(\frac{3}{2}\right)^{1/3} + \frac{1}{3}\left(\frac{2}{3}\right)^{2/3}\left(x - \frac{1}{2}\right) - \frac{1}{9}\left(\frac{2}{3}\right)^{5/3}\left(x - \frac{1}{2}\right)^2$$

Exercise 0.26 (Taylor expansion of $x \log x$)**.** Adapted from Miller (2016), Question 1.1.25.
Find the first three terms (up to degree 2) of the Taylor series for

$$f(x) = x \log x$$

at $x_0 = 1$.

Solution

Solution 0.26. Evaluate derivatives at $x_0 = 1$:

$$\begin{aligned} f(1) &= 1 \log 1 \\ &= 0 \\ f'(x) &= \log x + 1 \\ \implies f'(1) &= 0 + 1 \\ &= 1 \\ f''(x) = \frac{1}{x} &\implies f''(1) = 1 \end{aligned}$$

Therefore:

$$\begin{aligned} P_2(x) &= f(1) + f'(1)(x-1) + \frac{f''(1)}{2!}(x-1)^2 \\ &= (x-1) + \frac{1}{2}(x-1)^2 \end{aligned}$$

Exercise 0.27 (Maclaurin series of $\log(1+x)$)**.** Adapted from Miller (2016), Question 1.1.26.
Find the first three terms of the Taylor series for

$$f(x) = \log(1+x)$$

at $x_0 = 0$.

Solution

Solution 0.27. Differentiating repeatedly at $x_0 = 0$:

$$\begin{aligned} f(0) &= \log 1 \\ &= 0 \\ f'(x) = (1+x)^{-1} &\implies f'(0) = 1 \\ f''(x) = -(1+x)^{-2} &\implies f''(0) = -1 \\ f'''(x) = 2(1+x)^{-3} &\implies f'''(0) = 2 \end{aligned}$$

The Taylor polynomial of degree 3 is:

$$P_3(x) = x - \frac{x^2}{2} + \frac{x^3}{3}$$

Theorem 0.15 (Maclaurin polynomial of $\log(1+x)$)**.** *The Maclaurin polynomial (Definition 0.21) of degree 3 of* $f(x) = \log(1+x)$ *is*

$$P_3(x) = x - \frac{x^2}{2} + \frac{x^3}{3}$$

Proof

Proof. This is the solution to Exercise 0.27, worked in Solution 0.27. □

Exercise 0.28 (Maclaurin series of $\log(1-x)$)**.** Adapted from Miller (2016), Question 1.1.27.
Find the terms up to degree 3 of the Taylor series for

$$f(x) = \log(1-x)$$

at $x_0 = 0$.

Solution

Solution 0.28. Substitute $u = -x$ into the expansion for $\log(1+u)$ from Theorem 0.15:

$$\begin{aligned}\log(1-x) &= (-x) - \frac{(-x)^2}{2} + \frac{(-x)^3}{3} - \ldots \\ &= -x - \frac{x^2}{2} - \frac{x^3}{3} - \ldots\end{aligned}$$

Theorem 0.16 (Maclaurin polynomial of $\log(1-x)$)**.** *The Maclaurin polynomial (Definition 0.21) of degree 3 of* $f(x) = \log(1-x)$ *is*

$$P_3(x) = -x - \frac{x^2}{2} - \frac{x^3}{3}$$

Proof

Proof. This is the solution to Exercise 0.28, worked in Solution 0.28. □

Exercise 0.29 (Taylor series of $\log((1-x)e^x)$)**.** Adapted from Miller (2016), Question 1.1.28.
Find the first two non-zero terms of the Taylor series for

$$f(x) = \log((1-x)e^x)$$

at $x_0 = 0$.

Solution

Solution 0.29. Expand using logarithm properties:

$$\begin{aligned} f(x) &= \log(1-x) + \log(e^x) \\ &= \log(1-x) + x \end{aligned}$$

Substitute the series for $\log(1-x)$ from Theorem 0.16:

$$\begin{aligned} f(x) &= \left(-x - \frac{x^2}{2} - \frac{x^3}{3} - ...\right) + x \\ &= -\frac{x^2}{2} - \frac{x^3}{3} - ... \end{aligned}$$

The constant and linear terms vanish ($f(0) = 0$, $f'(0) = 0$). The first two non-zero terms are:

$$-\frac{x^2}{2} - \frac{x^3}{3}$$

Exercise 0.30 (Taylor series of a product of known expansions)**.** Adapted from Miller (2016), Question 1.1.29.
Find the terms up to degree 2 of the Taylor series for

$$f(x) = \cos(x)\log(1+x)$$

at $x_0 = 0$.

Solution

Solution 0.30. Multiply the known Maclaurin expansions:

$$\begin{aligned} \cos x &= 1 - \frac{x^2}{2} + O(x^4) \\ \log(1+x) &= x - \frac{x^2}{2} + O(x^3) \end{aligned}$$

Multiplying and dropping terms of order x^3 and higher:

$$
\begin{aligned}
f(x) &= \left(1 - \frac{x^2}{2} + ...\right)\left(x - \frac{x^2}{2} + ...\right) \\
&= x - \frac{x^2}{2} + O(x^3)
\end{aligned}
$$

Exercise 0.31 (Taylor series of $\log(1 + 2x)$)**.** Adapted from Miller (2016), Question 1.1.30.
Find the first two non-zero terms of the Taylor series for

$$f(x) = \log(1 + 2x)$$

at $x_0 = 0$.

Solution

Solution 0.31. Substitute $u = 2x$ into the Maclaurin series $\log(1 + u) = u - \frac{u^2}{2} + ...$:

$$
\begin{aligned}
\log(1 + 2x) &= (2x) - \frac{(2x)^2}{2} + ... \\
&= 2x - 2x^2 + ...
\end{aligned}
$$

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